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K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

End Semester Exam**April-May 2016****Duration: 3 Hrs****Semester: IV****Branch: COMP/IT****KJSC 2014****Max. Marks: 100****Class: S Y BTech****Name of the Course: Applied Mathematics-IV****Course Code: UCEC401 / UIRC401****Instructions:****(1) All Questions are Compulsory****(2) Figures to right indicate full marks. Each sub-question has equal marks.**

Question NO.	Questions	Marks																						
Q.1	Attempt any THREE of the following.	7X3=21 Marks																						
	(i) Find the coefficient of Rank correlation for the following data <table><tr><td>X</td><td>32</td><td>55</td><td>49</td><td>60</td><td>43</td><td>37</td><td>43</td><td>49</td><td>10</td><td>20</td></tr><tr><td>Y</td><td>40</td><td>30</td><td>70</td><td>20</td><td>30</td><td>50</td><td>72</td><td>60</td><td>45</td><td>25</td></tr></table>		X	32	55	49	60	43	37	43	49	10	20	Y	40	30	70	20	30	50	72	60	45	25
	X		32	55	49	60	43	37	43	49	10	20												
	Y		40	30	70	20	30	50	72	60	45	25												
	(ii) obtain equation of regression line of x on y. Also find value of y when x = 60. <table><tr><td>X</td><td>78</td><td>36</td><td>98</td><td>25</td><td>75</td><td>82</td><td>90</td><td>62</td><td>65</td><td>39</td></tr><tr><td>Y</td><td>84</td><td>51</td><td>91</td><td>60</td><td>68</td><td>62</td><td>86</td><td>58</td><td>53</td><td>47</td></tr></table>		X	78	36	98	25	75	82	90	62	65	39	Y	84	51	91	60	68	62	86	58	53	47
X	78	36	98	25	75	82	90	62	65	39														
Y	84	51	91	60	68	62	86	58	53	47														
(iii) From a partially destroyed laboratory record only following are available. $x = 4y + 5$ and $y = Kx + 4$ are regression lines of x on y and y on x respectively. show that $0 \leq K \leq \frac{1}{4}$. If $K = \frac{1}{16}$ find the means of the two variables x and y and also correlation coefficient between them																								
(iv) Fit a second degree curve to the following data taking x as the independent variable. <table><tr><td>year (X)</td><td>1974</td><td>1975</td><td>1976</td><td>1977</td><td>1978</td><td>1979</td><td>1980</td><td>1981</td></tr><tr><td>Production (Y)</td><td>12</td><td>14</td><td>26</td><td>42</td><td>40</td><td>50</td><td>52</td><td>53</td></tr></table>	year (X)	1974	1975	1976	1977	1978	1979	1980	1981	Production (Y)	12	14	26	42	40	50	52	53						
year (X)	1974	1975	1976	1977	1978	1979	1980	1981																
Production (Y)	12	14	26	42	40	50	52	53																
Q.2	Attempt any THREE of the following.	7X3=21 Marks																						
	(i) A continuous random variable X has the p.d.f. defined by $f(x)=A+Bx$, $0 \leq x \leq 1$. If the mean of the distribution is $1/3$, find A, B and variance.																							
	(ii) If m things are distributed among 'a' men & 'b' women, show that the probability that the number of things received by men is odd is $\frac{(b+a)^m - (b-a)^m}{2(b+a)^m}$																							
	(iii) In an intelligence test administered to 1000 students the average was 42 and standard deviation was 24. Find the number of students (i) exceeding 50 (ii) between 30 and 54, and (iii) the least score of top 100 students.																							
	(iv) If X is normal variate with mean 25 and standard deviation 5. Find the C such that $P(X \leq C) = 0.01$ and $P(X \geq C) = 0.05$																							
(v) A Manufacturer of electric bulbs sends out 500 lots each consisting of 100 bulbs. If 5% bulbs are defective, then in how many lots can we expect (i) 97 or more good bulbs (ii) less than 96 good bulbs?																								
Q.3	Attempt any THREE of the following.	7X3=21 Marks																						
	(i) The means of two samples of sizes 1000 and 2000 respectively are 67.50 and 68 inches. Can the samples be regarded as drawn from the same population of standard deviation 2.5 inches? (Take 5% level of significance)																							

	(ii) Table below shows the performances of student in mathematics and physics. Using Chi-square test the hypothesis that the performance in mathematics is independent of performance in physics. (Take 5% level of significance) <table><tr><td></td><td colspan="4">Grades in Mathematics</td></tr><tr><td></td><td></td><td>High</td><td>Medium</td><td>Low</td></tr><tr><td rowspan="3">Grades in physics</td><td>High</td><td>56</td><td>71</td><td>12</td></tr><tr><td>Medium</td><td>47</td><td>163</td><td>38</td></tr><tr><td>Low</td><td>14</td><td>42</td><td>81</td></tr></table>		Grades in Mathematics						High	Medium	Low	Grades in physics	High	56	71	12	Medium	47	163	38	Low	14	42	81	
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		High	Medium	Low																					
Grades in physics	High	56	71	12																					
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	Low	14	42	81																					
	(iii) Two independent samples of sizes 8 & 7 contained the following values <table><tr><td>Sample I</td><td>19</td><td>17</td><td>15</td><td>21</td><td>16</td><td>18</td><td>16</td><td>14</td></tr><tr><td>Sample II</td><td>15</td><td>14</td><td>15</td><td>19</td><td>15</td><td>18</td><td>16</td><td></td></tr></table> Is the difference between sample means significant ?(Take 5% level of significance)	Sample I	19	17	15	21	16	18	16	14	Sample II	15	14	15	19	15	18	16							
Sample I	19	17	15	21	16	18	16	14																	
Sample II	15	14	15	19	15	18	16																		
	(iv) The number of defects in printed circuit board is hypothesised to follow the poisson distribution. A random sample of 60 printed boards showed the following data. <table><tr><td>Number of defects</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>Observed frequency</td><td>32</td><td>15</td><td>9</td><td>4</td></tr></table> Does the hypothesis of Poisson distribution seem appropriate.(Take 5% level of significance)	Number of defects	0	1	2	3	Observed frequency	32	15	9	4														
Number of defects	0	1	2	3																					
Observed frequency	32	15	9	4																					
Q.4	Attempt any TWO of the following. (i) In a railways marshalling yard, good trains arrive at a rate of 30 trains per day. Assuming that the interarrival times follows an Piosson distribution and the service time (the time taken to hump a train) distribution is also exponential with an average of 36 minutes. Calculate. (a) expected number of train in the system. (b) probability that the queue size greater than equal 10 If the input of trains increases to an average of 33 per day, whar will be the change in (a) and (b)? (ii) Consider a self service store with one cashier. Assume Piosson arrivals and exponential serice times. Suppose that on average nine customers arrive every 5 minutes and that the cashier can serve 10 in 5 minutes. Find: (a) average number of customers in the system. (b) probability of having more than 10 customers in system and (c) probability that a customer has to queue for more than 2 minutes. (iii) Consider a single serve queuing system with Poisson input and exponential service times. Suppose the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hour and the maximum permissible calling units in the system is two. Derive the steady- state probability distribution of the number of calling units in the system, then calculate the expected number in system and also find traffic intensity.	8X2=16 Marks																							
Q.5	Attempt any THREE of the following. (i) Solve the following N.L.P.P. Maximise $z = 8x_1 + 10x_2 - x_1^2 - x_2^2$ Subject to $3x_1 + 2x_2 \leq 6$ $x_1, x_2 \geq 0$	7X3=21 Marks																							

(ii)	Find (i) All basic solutions (ii) All feasible basic solutions (iii) Optimal feasible basic solution for the following L.P.P. maximise $z = 2x_1 - 2x_2 + 4x_3 - 5x_4$ Subject to $x_1 + 4x_2 - 2x_3 + 8x_4 = 2$ $-x_1 + 2x_2 + 3x_3 + 4x_4 = 1$ where $x_1, x_2, x_3, x_4 \geq 0$	
(iii)	Solve the following L.P.P. by Big M method maximise $z = 3x_1 - x_2$ Subject to $2x_1 + x_2 \leq 2$ $x_1 + 3x_2 \geq 3$ $x_2 \leq 4$ where $x_1 \geq 0, x_2 \geq 0$	
(iv)	Using the method of Lagrange's Multiplier solve the following N.L.P.P. Optimize $z = 12x_1 + 8x_2 + 6x_3 - x_1^2 - x_2^2 - x_3^2 - 23$ Subject to $x_1 + x_2 + x_3 = 10$ where $x_1, x_2, x_3 \geq 0$	